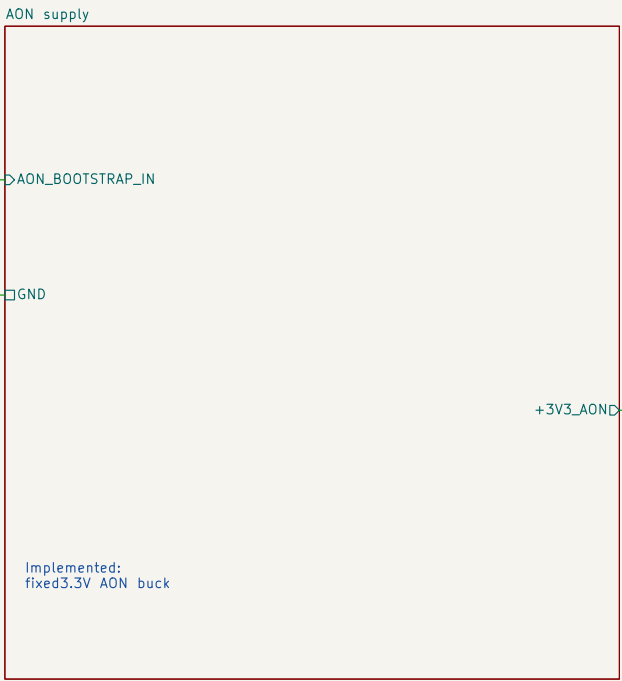
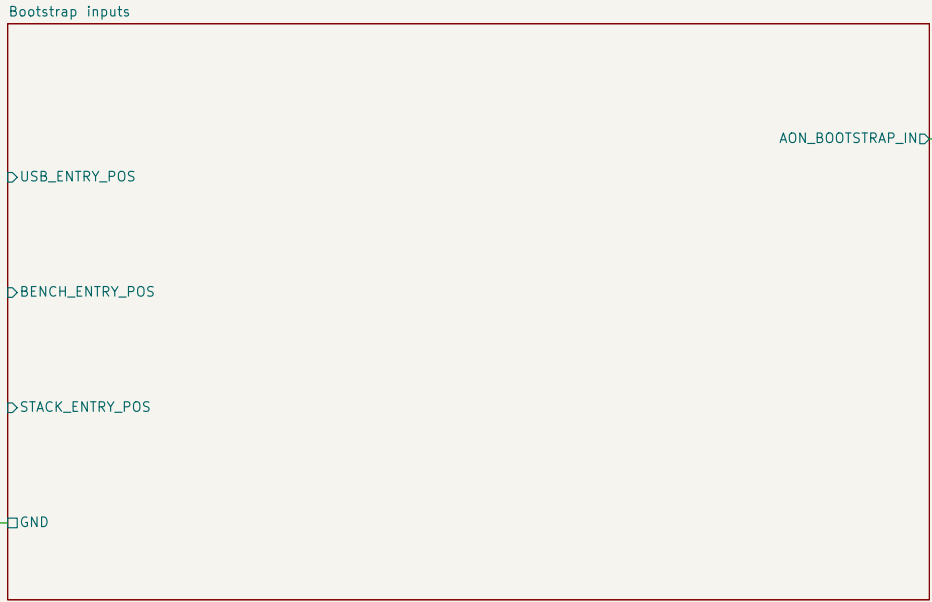


EMBER PAYLOAD COMPUTE – NEXT PROTOTYPE / STAGE 3



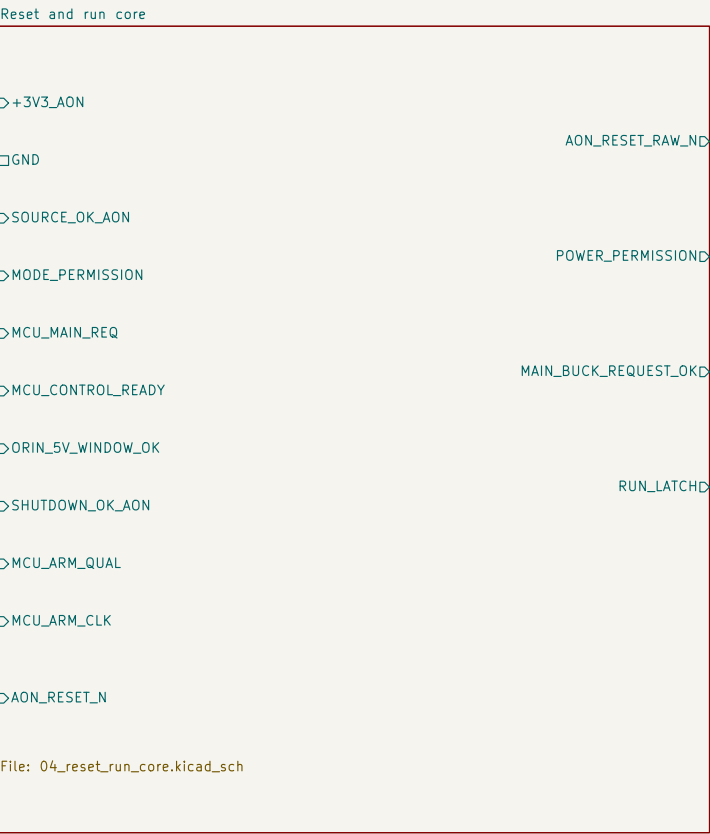
File: 03_aon_supply.kicad_sch



File: 02_bootstrap_inputs.kicad_sch

Implemented: three autonomous limited / reverse-blocked bootstrap paths.
USB Initia5V; bench6-8.4V; stack range still bounded by entry contract.
Sources below are requests, not attached-source voltage proof.

OPEN BOUNDARIES / FAIL-OFF CORE
SOURCE_OK_AON: raw windows, source presence, PD contract,
selected bus, inrush and fault qualification are not captured.
ORIN_5V_WINDOW_OK and SHUTDOWN_OK_AON remain qualified receiver inputs.
MCU requests, arm, clocks, ready and WDI need future STM32.
Bench jumper, IHU and MCU/debug connector hardware still pending.
MCU_NRST_N shared watchdog output accepts only external OD drivers.

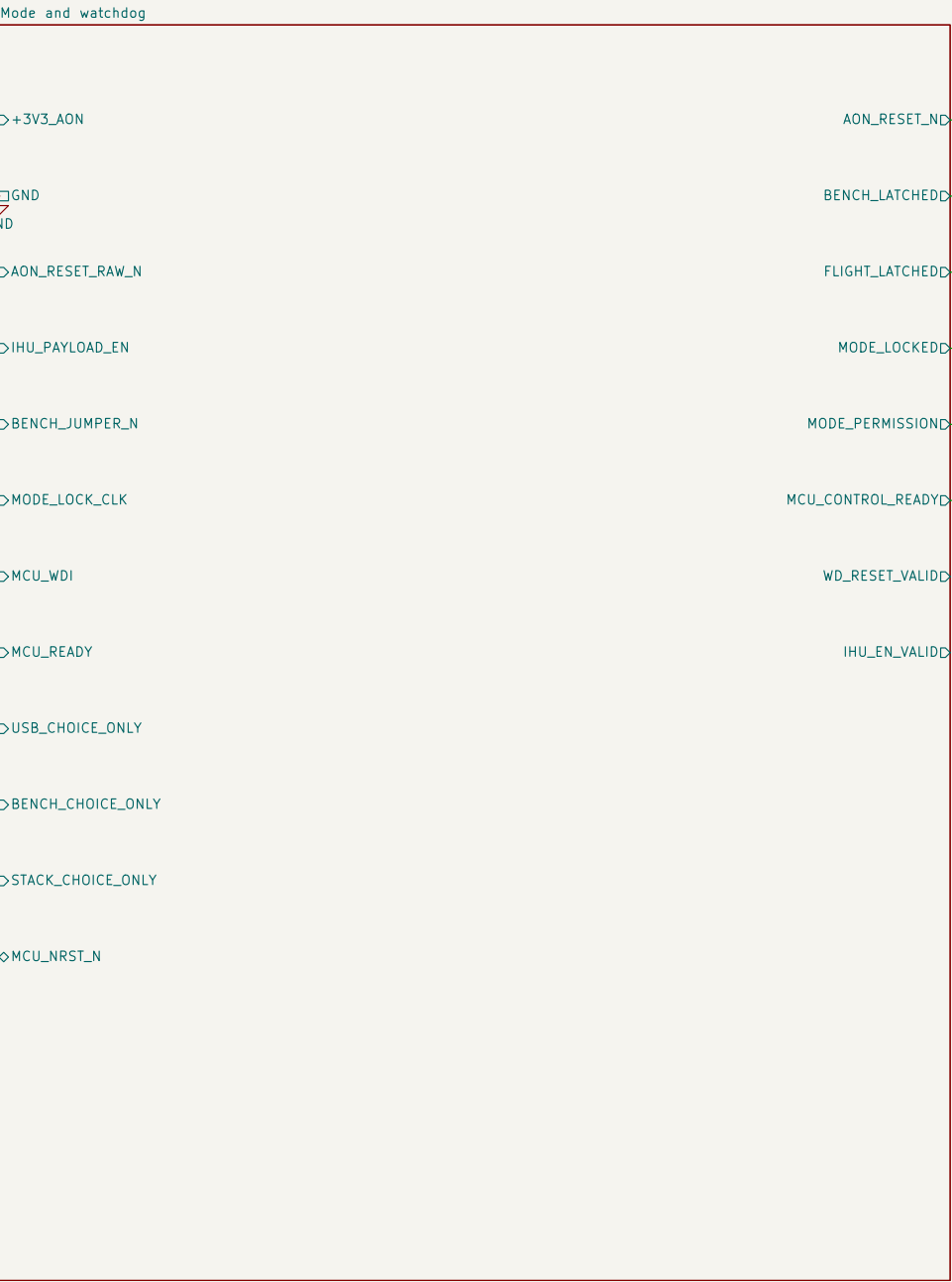


AON_RESET_RAW_NC

POWER_PERMISSIOND

MAIN_BUCK_REQUEST_OKD

RUN_LATCHD



AON_RESET_NC

BENCH_LATCHEDD

FLIGHT_LATCHEDD

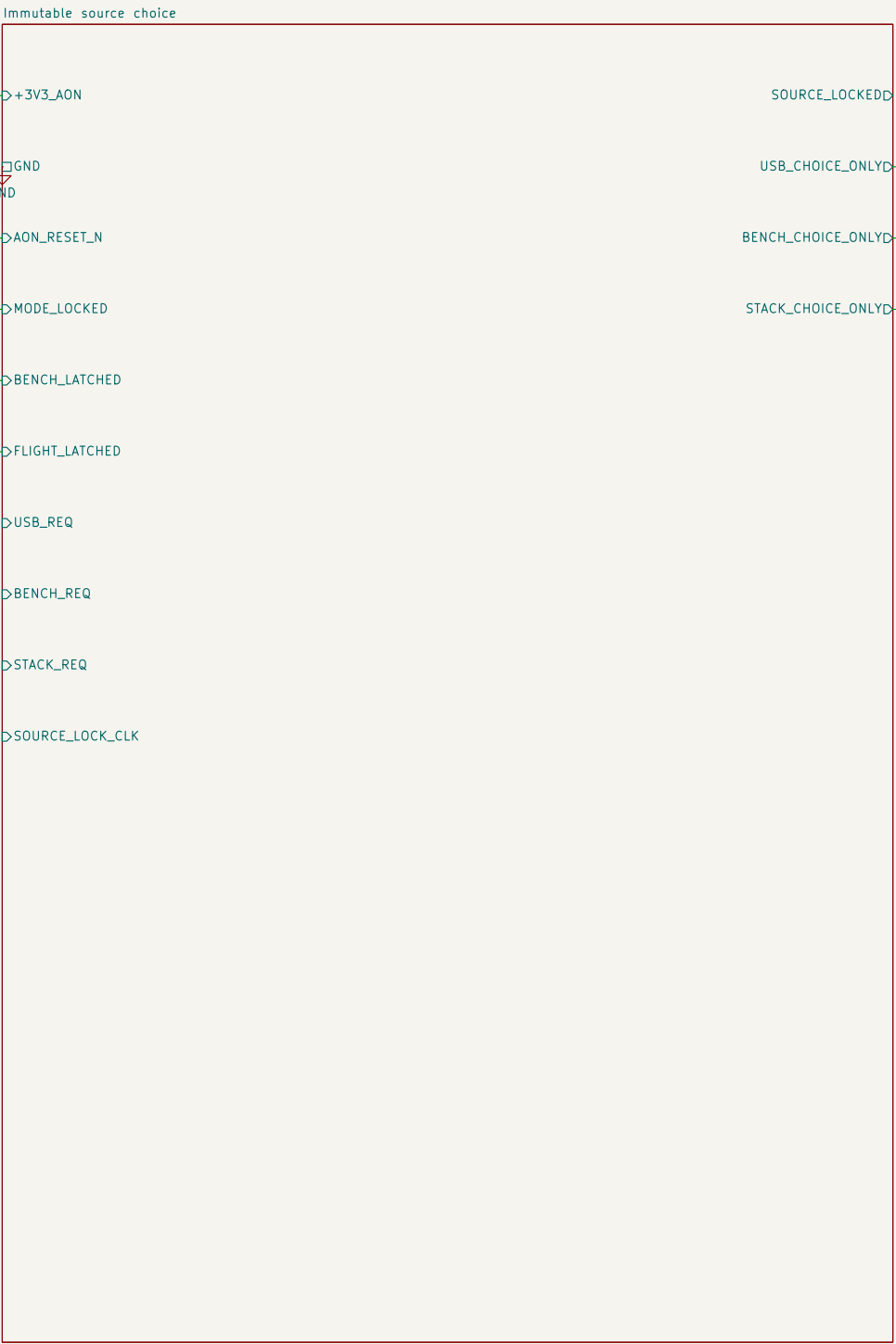
MODE_LOCKEDD

MODE_PERMISSIOND

MCU_CONTROL_READYD

WD_RESET_VALIDD

IHU_EN_VALIDD



File: 06_source_lock.kicad_sch

IMPLEMENTED HARDWARE OWNERSHIP
Mode startup latch / source-lock latch / one-hot choice.
MODE_PERMISSION requires mode-compatible locked exclusive choice;
Flight also requires live IHU_PAYLOAD_EN high.
MCU_CONTROL_READY includes WDI enabled/healthy, reset and MCU_READY.
SOURCE_OK remains independent actual power qualification.

PROTOTYPE QUALIFICATION GATES
AON startup target0.200W / ceiling0.250W; active0.350W deferred.
New logic budget is conditional: cold buck ramp/inrush still unproven.
Schmitt OD receiver thresholds and edges need full rail/ramp measurement.
Mode waits20ms reset release +10ms stable: lock1ms only once per AON cycle.
Source request setup/hold1ms around1ms lock pulse.
No layout, fabrication, assembly or flight release.

Sheet: /
File: payload_compute_rev_next.kicad_sch

Title:

Size: A2

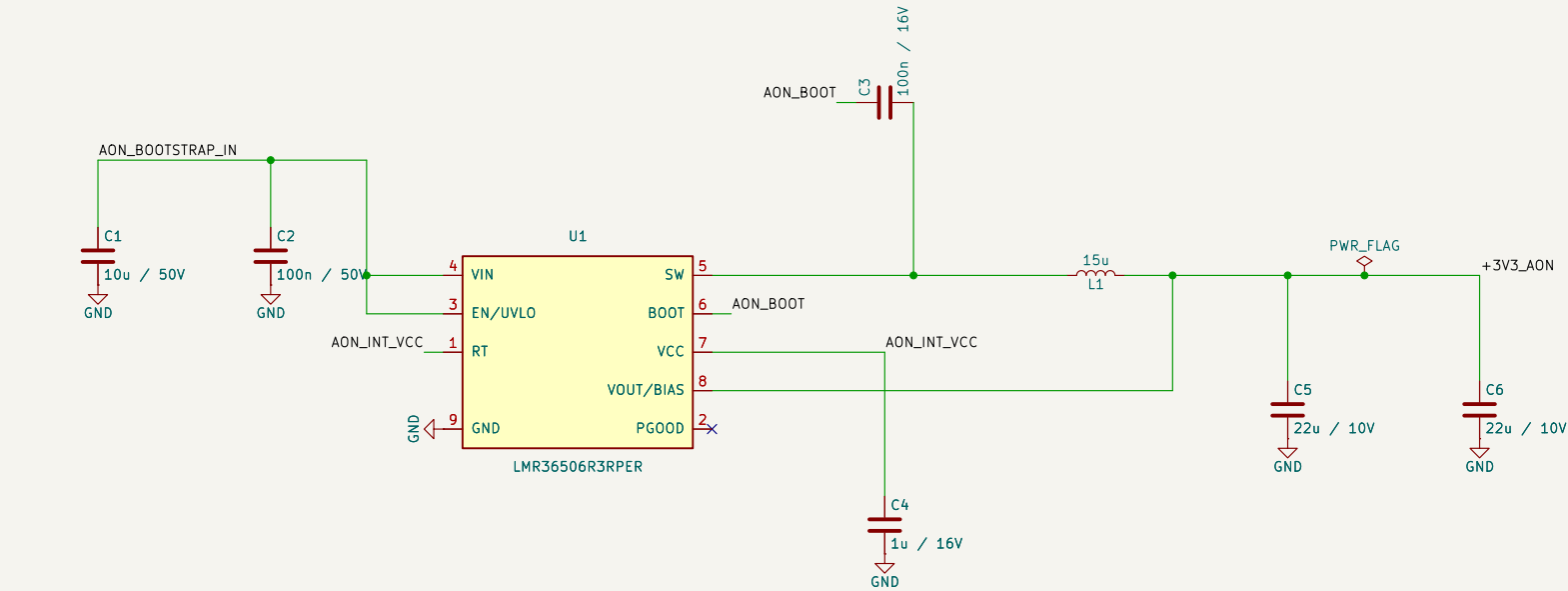
Date:

Rev:

KiCad E.D.A. 10.0.0

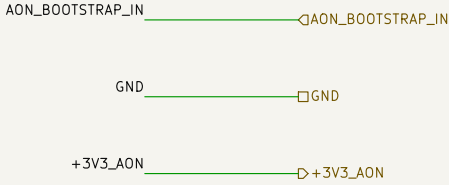
Id: 6/6

03 – ALWAYS-ON SUPPLY / IMPLEMENTED CORE

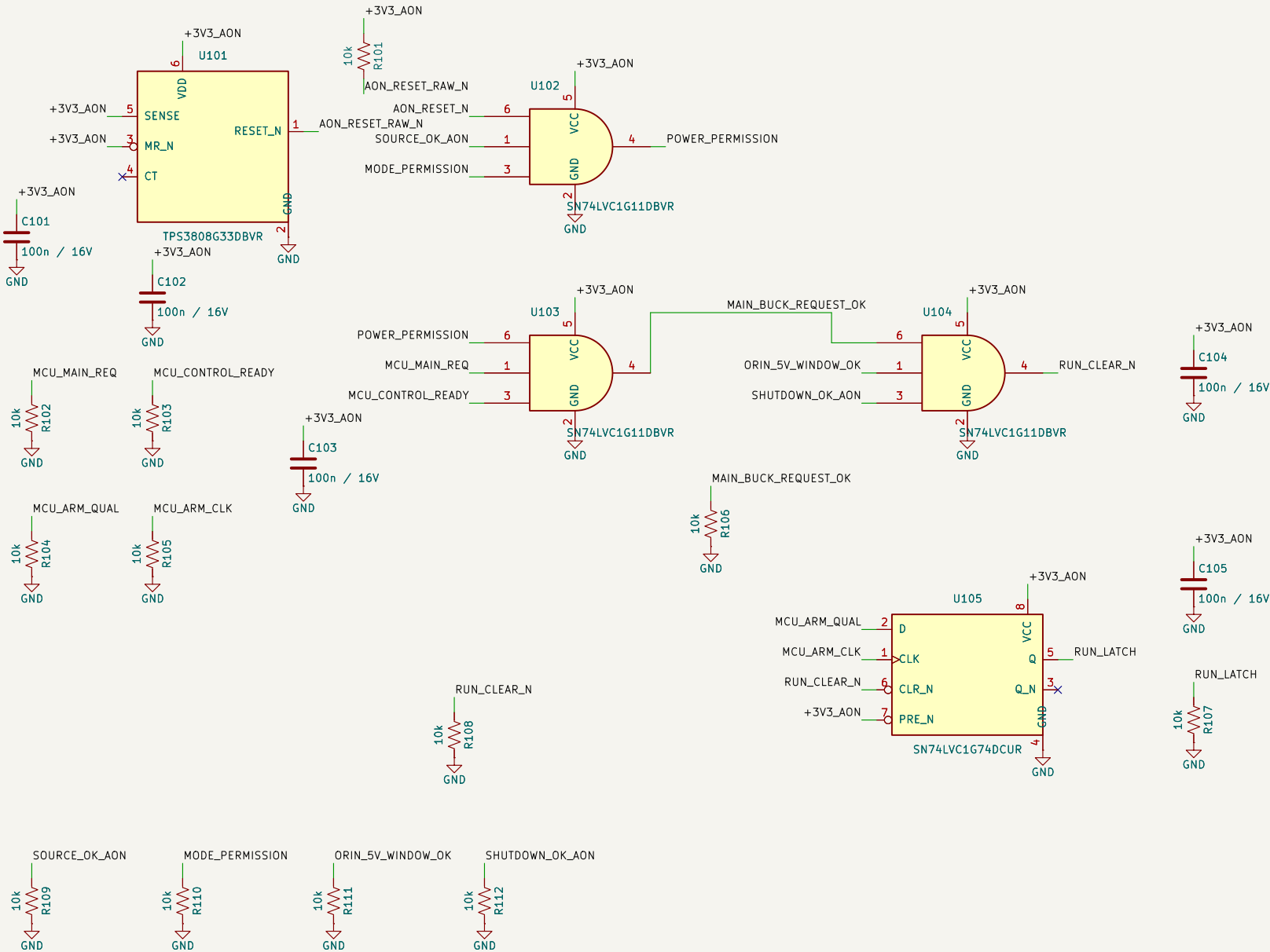


RT tied to internal VCC; nominal 1 MHz. VCC supplies only RT selection and C4; no external load.
15 uH, 2 x 22 uF output are nominal prototype values; effective C / saturation / ripple / MPN remain to qualify.
PGOOD unused. AON reset is supervised separately by TPS3808G33. EN follows protected bootstrap feed.

INTERFACE / ASSEMBLY GATES
Limited native-OR bootstrap feeds are captured. AON_BOOTSTRAP_IN is not raw stack or unprotected USB.
Coordinated entry protection and cold-start/ramp qualification remain pending.
Chosen fixed-output DC allocation: 3.2505-3.3825 V; max G33 release 3.1928 V; static margin 57.7 mV.
Startup budget target 0.200 W (old ceiling 0.250 W); transients, lower-feed startup and attach current unmeasured.
PWR_FLAG on generated +3V3_AON declares the real buck output through L1; no flag declares missing input/return.
Accepted RPE prototype lands differ from TI contour; common 0.100 / 0.125 mm stencil still to reconcile.

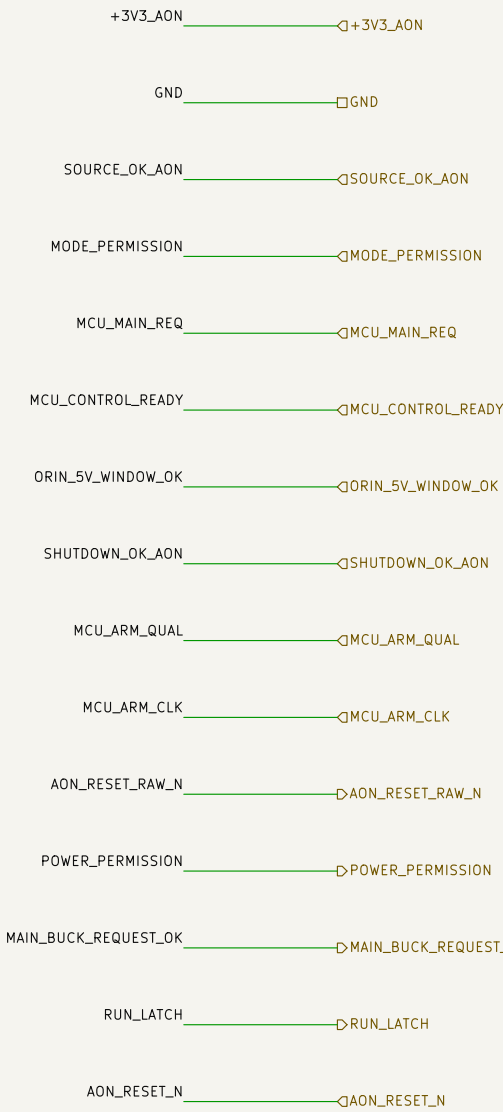


04 – AON RESET / PERMISSION / FRESH-ARM RUN CORE



POWER_PERMISSION = AON_RESET_N & SOURCE_OK_AON & MODE_PERMISSION
MAIN_BUCK_REQUEST_OK = POWER_PERMISSION & MCU_MAIN_REQ & MCU_CONTROL_READY
RUN_CLEAR_N = MAIN_BUCK_REQUEST_OK & ORIN_5V_WINDOW_OK & SHUTDOWN_OK_AON
RUN_LATCH samples MCU_ARM_QUAL on rising MCU_ARM_CLK only while RUN_CLEAR_N is high.
Fault/recovery does not generate an arm edge. PRE_N tied high; Q_N and TPS3808 CT intentionally NC.

BOUNDARY / TIMING CONTRACT
All external high-true qualifiers and MCU controls have 10k pulldowns (about 0.342 mA worst at 3.3825 V / 9.9k).
Future drivers must meet LVC VIH >= 2.0 V / VIL <= 0.8 V with these loads, ramps and partial power checked.
CT open: AON release 12-28 ms; not source/reference/module readiness. MR_N tied high.
Arm: keep CLK low, set D, wait >= 1 ms, pulse CLK >= 1 ms, lower CLK then D. Race/recovery tests remain.
Logic request is not physical buck EN. RUN_LATCH is not direct 5 V Jetson POWER_EN.



Sheet: /Reset and run core/
File: 04_reset_run_core.kicad_sch

Title:

Size: A3

Date:

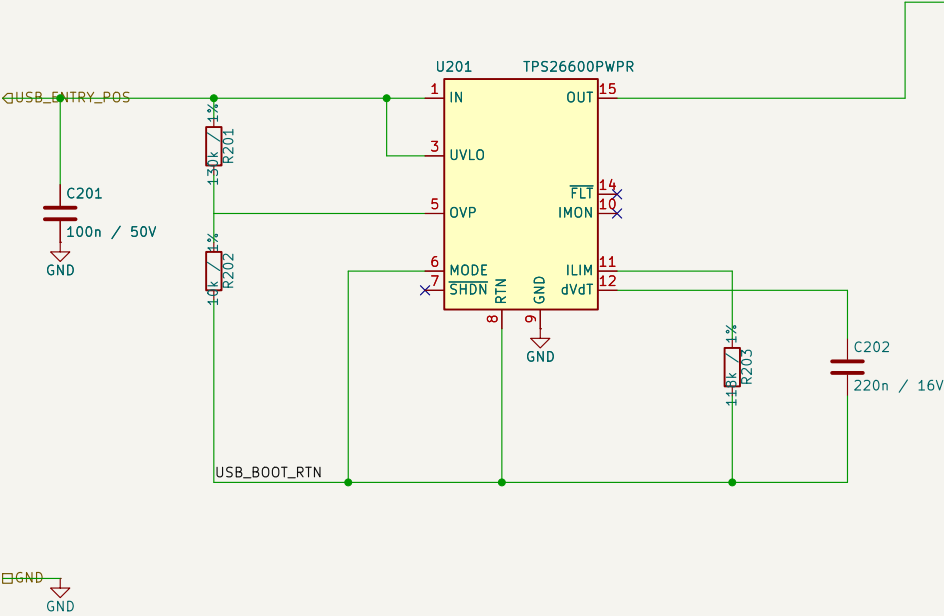
Rev:

KiCad E.D.A. 10.0.0

Id: 3/6

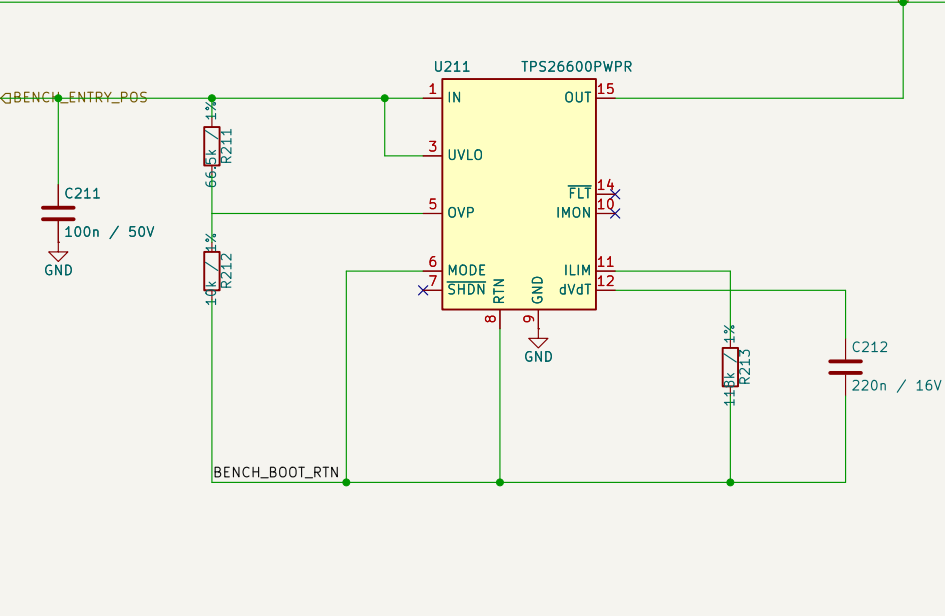
USB autonomous bootstrap

C-to-C >=500mA at initial5V
5V attach / 15V PD operating



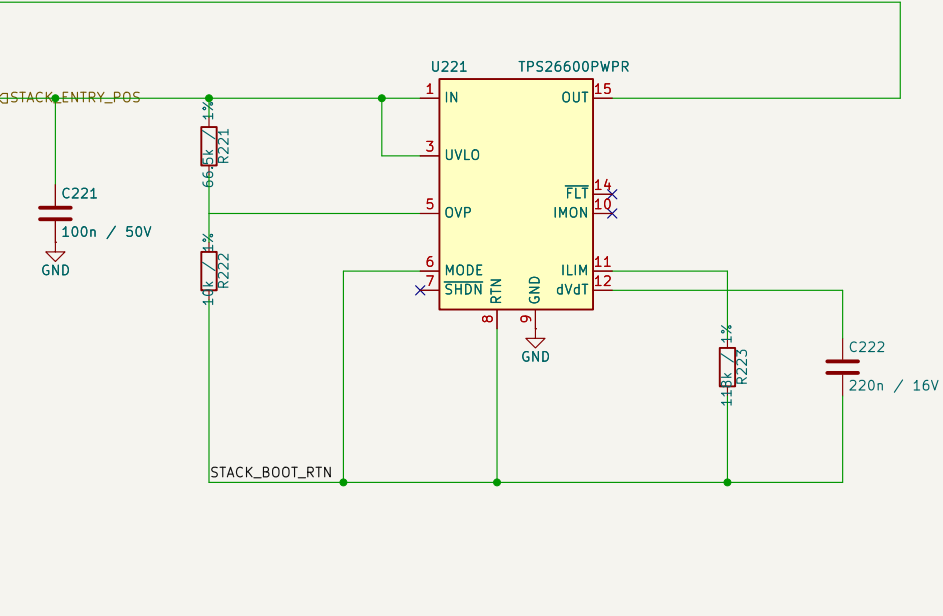
BENCH autonomous bootstrap

6-8.4V normal source
Entry fault envelope provisional



STACK autonomous bootstrap

6-8.4V normal source
Entry fault envelope provisional

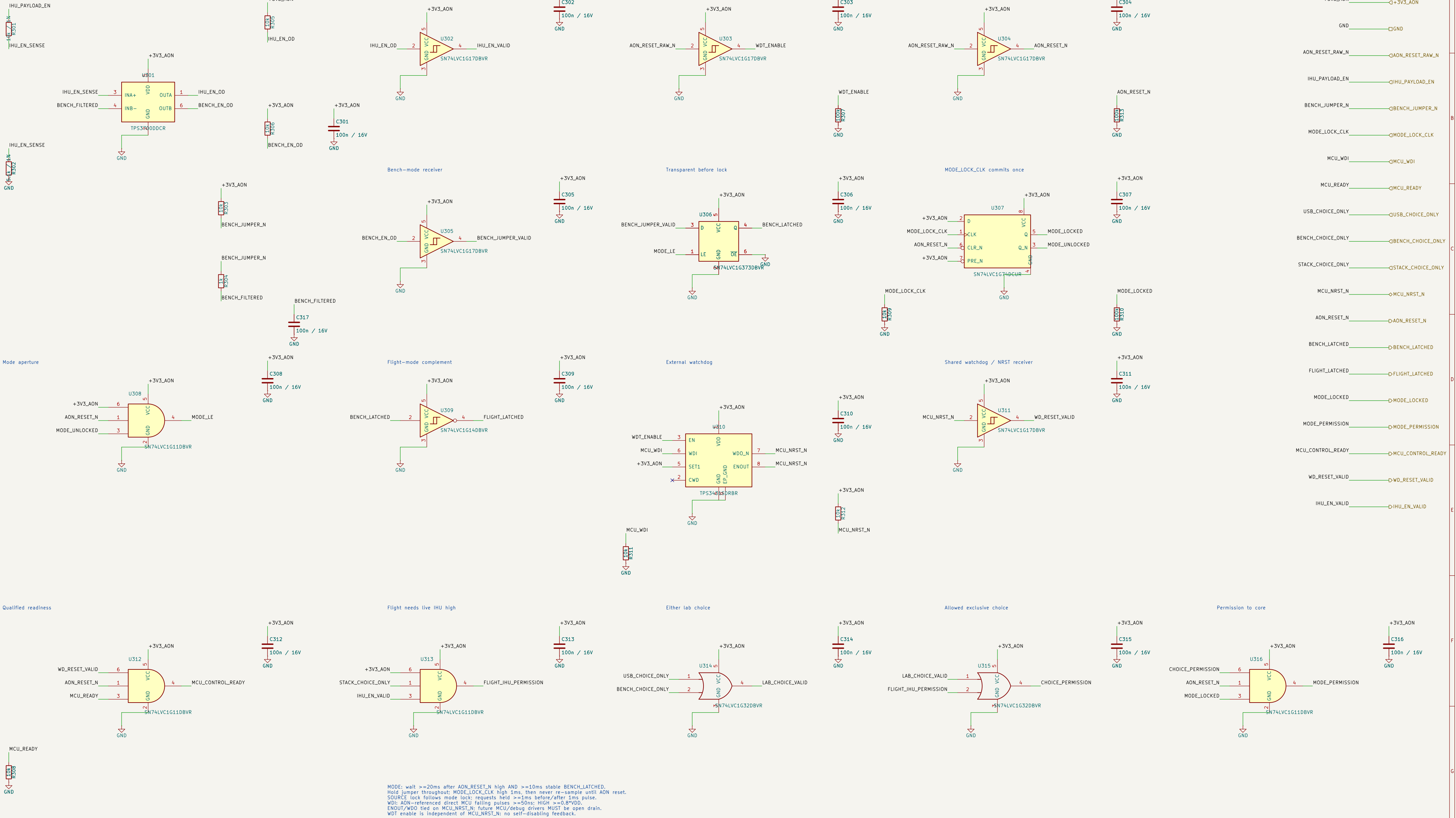


Stage2: autonomous limited AON paths; main admissions still absent.
USB UVLO is tied directly to IN to allow initial5V, not factory15V.
SHDN self-biased open; MODE=localRTN gives active limit/auto-retry.
All three RTN/EP domains remain separate from GND and one another.
Native reverse-blocking OR replaces Schottky; leakage/transients need tests.
RILIM118k/1% stays below120k recommendation; short limit ~123mA max.
Cin100nF each; shared switched AON input has10uF+100nF on supply sheet.
Reduced AON0.200W target /0.250W ceiling; CAN standby until full profile.
Source connector, PD, coordinated entry fuse/TVS and flight inhibit are boundaries.
Inputs provisional<=24V fault envelope; no60V system rating claimed.
Cold attach, slow ramp with VIN-enabled buck, restart/leakage remain scope gates.

CdVdt220nF / 20% allocation; output slew ~360-797V/s.
Charging12.12uF switched input capacitance needs <=9.66mA at ramp end.
This arithmetic does not bound initial buck/output-cap charging load.
Bootstrap OR is independent of main immutable source selection.
Fault retry or complete AON loss must never synthesize a new arm.

MODE / IHU / WATCHDOG - PROTOTYPE CAPTURE

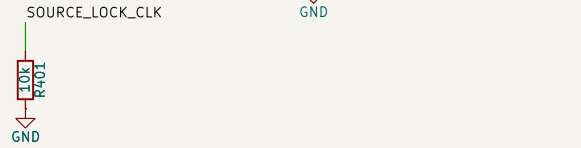
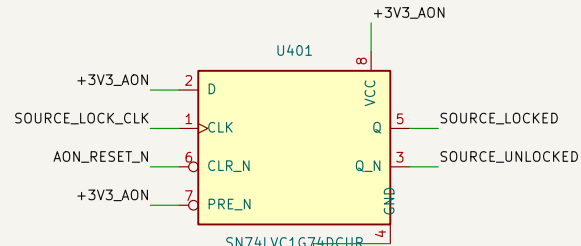
Grounded startup jumper selects bench; open selects flight.



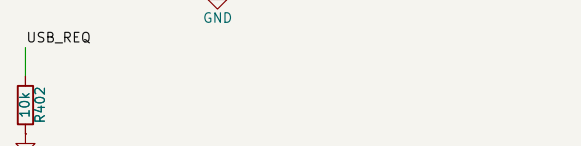
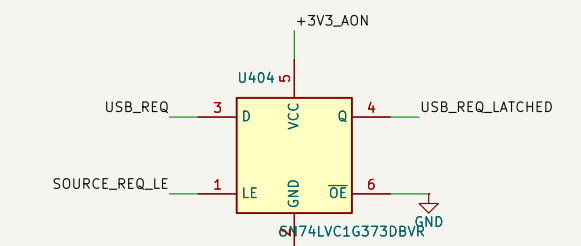
Near-rail OD receivers use Schmitt LVC as a CONDITIONAL prototype.
VT limits are discrete VCC tests: full AON-ramp/edge behavior must be measured.
WDT_ENABLE (load limited to EN + 100k pulldown (<100uA)).
SOURCE_OK, raw presence and main source admission remain future hardware.

IMMUTABLE SOURCE REQUEST / ONE-HOT CHOICE

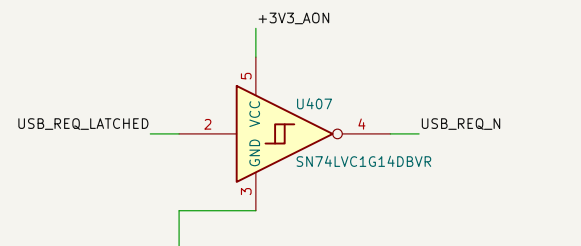
Only AON reset clears source lock



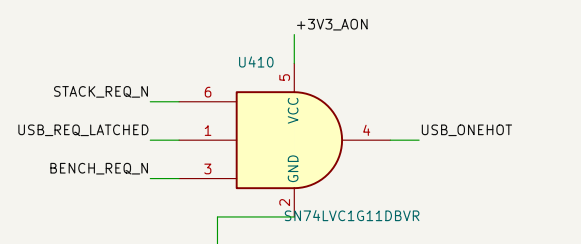
USB request latch



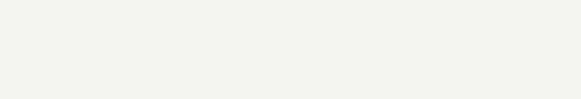
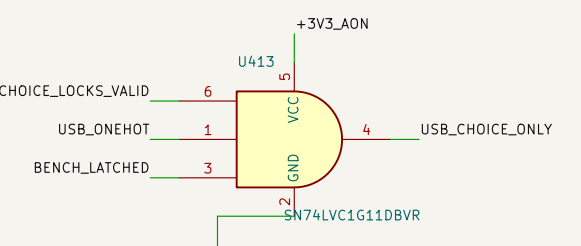
Complement



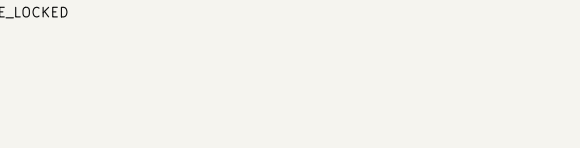
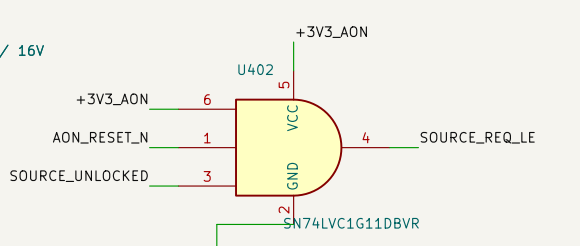
USB only (100)



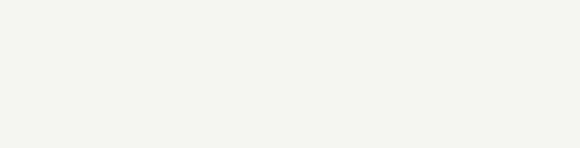
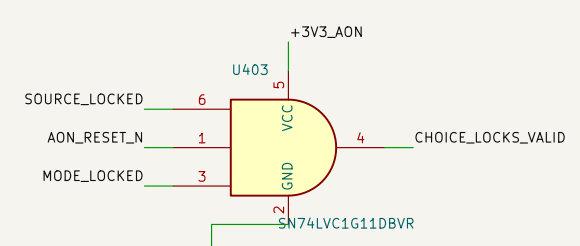
USB requires bench mode



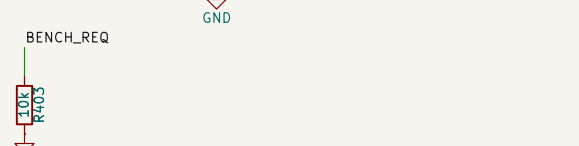
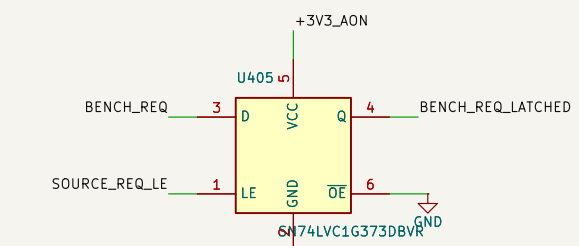
Request latch aperture



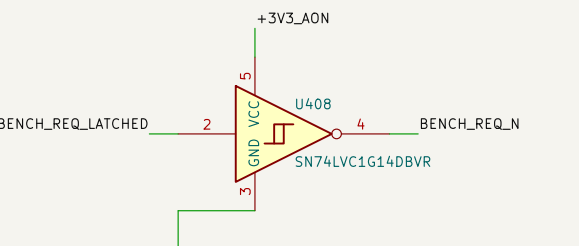
Both startup locks required



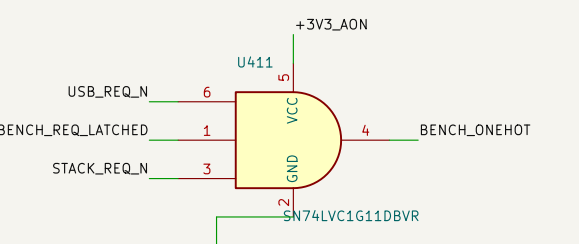
Bench request latch



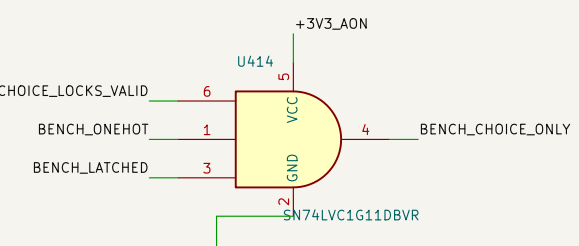
Complement



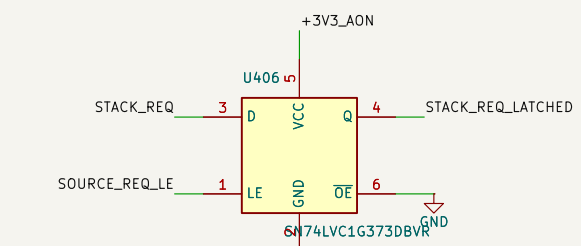
Bench only (010)



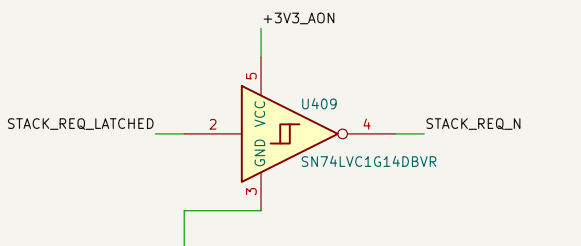
Bench requires bench mode



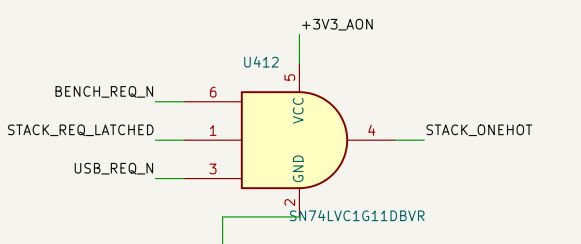
Stack request latch



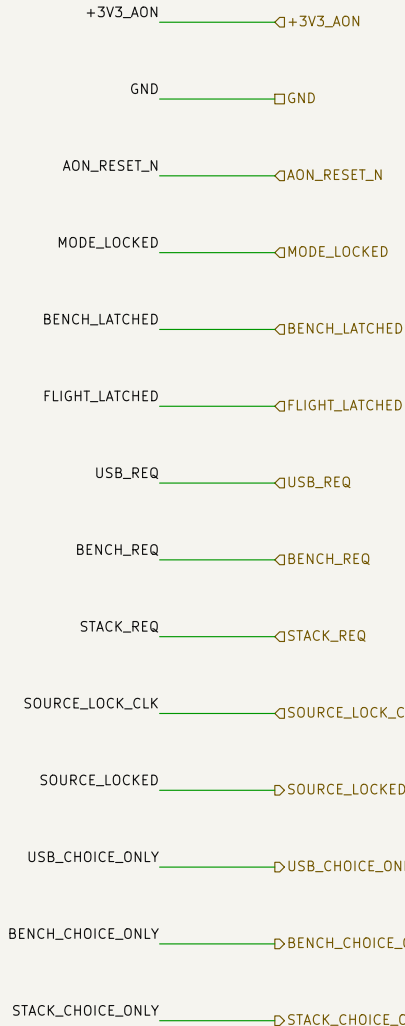
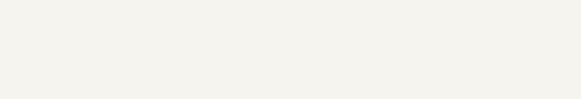
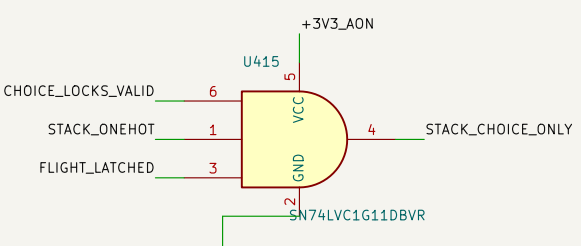
Complement



Stack only (001)



Stack requires flight mode



000 and all multihot requests produce no choice.
These outputs identify a locked request; they do NOT prove source voltage, PD contract, raw attachment exclusivity, fault status, selected bus or inrush completion.
No hardware retarget after ordinary MCU reset: AON reset alone reopens the aperture.
Firmware sets requests >=1ms before/after the first SOURCE_LOCK_CLK 1ms pulse.
All later lock pulses cannot change the stored request.